GOSIM®
GLOBAL OPEN-SOURCE INNOVATION MEETUP

GOSIM Paris 2026

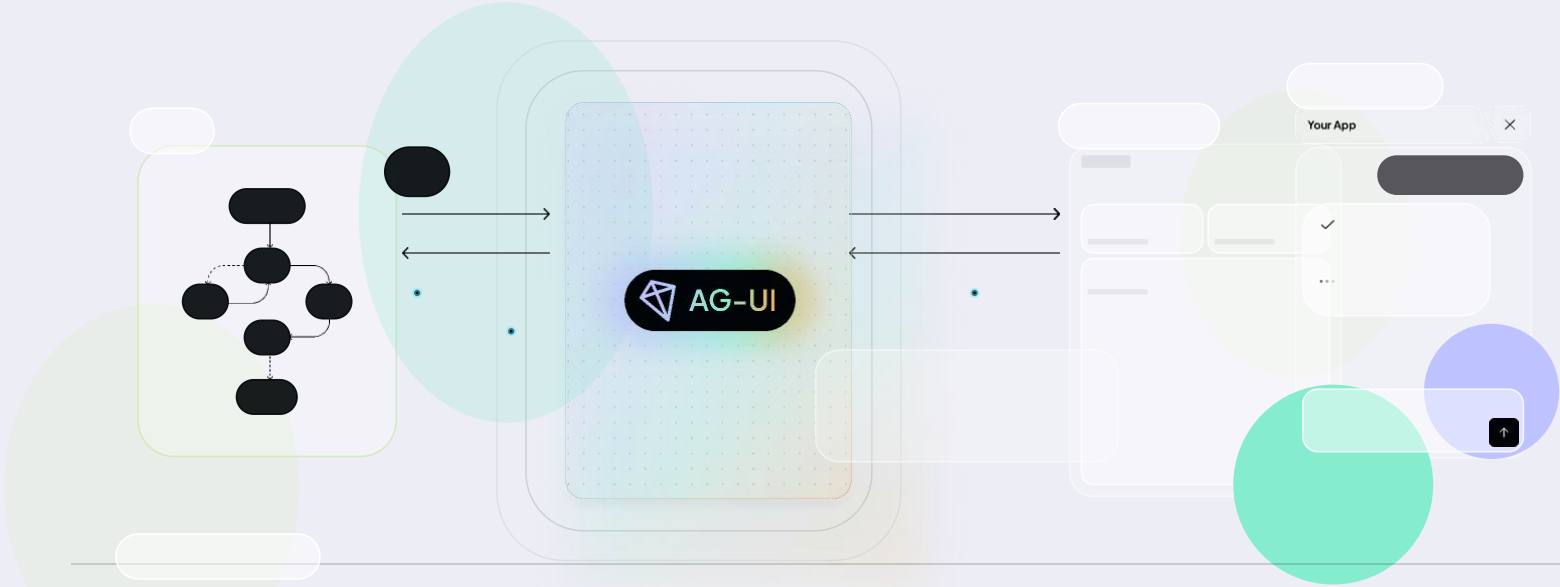
Topic

Building Agentic Apps with CopilotKit and AG-UI

Ran Shemtov

Founding Engineer, CopilotKit

CopilotKit



mastra Pydantic AI LangGraph Agno AG2 crewai LlamaIndex

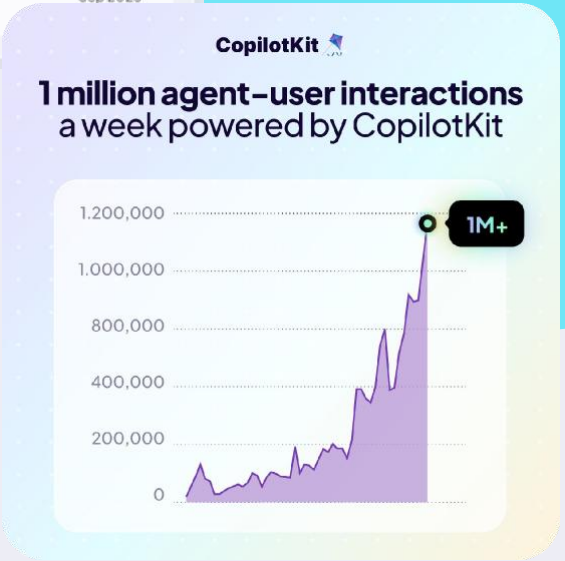
aws

CopilotKit

-
-
-
-
-


Weekly Installs
AG-UI
CopilotKit
Growing 24% MoM
Total: 120K+
80k
60k
40k
20k
0
Oct 2024
Sep 2025

CopilotKit

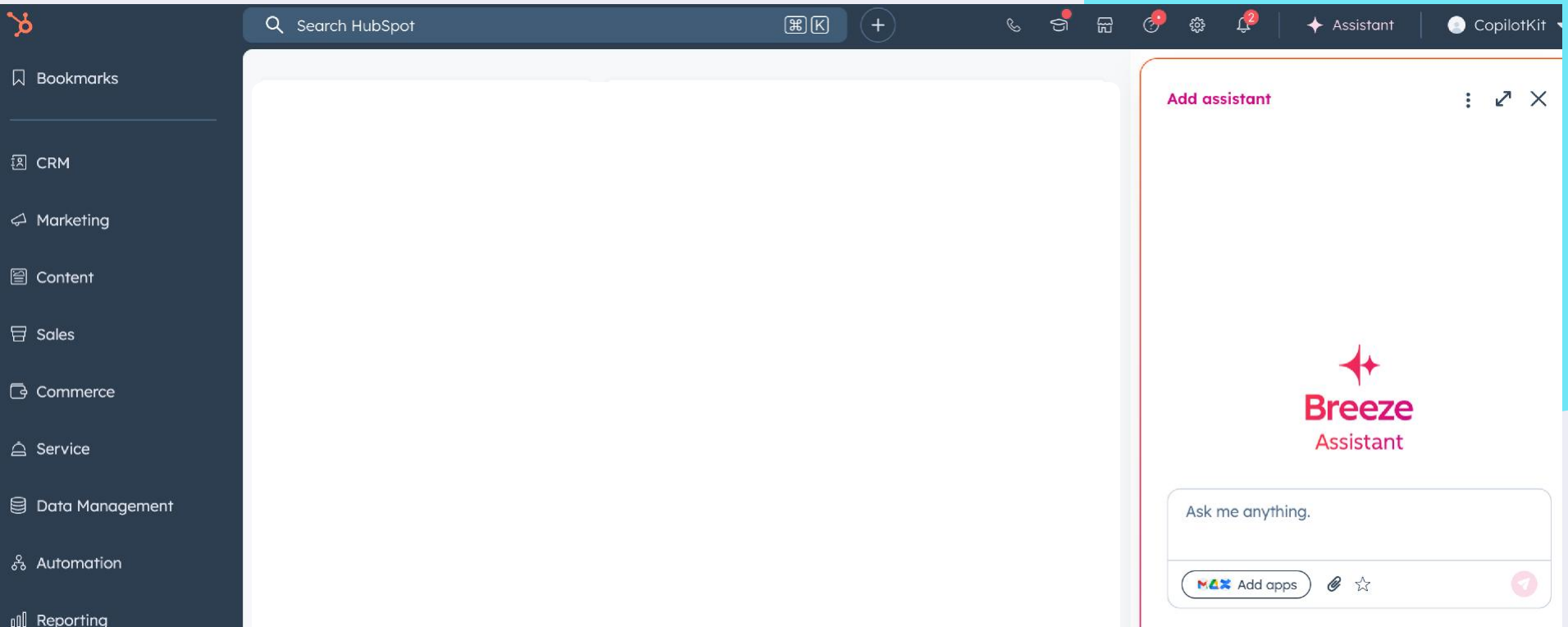

CopilotKit
1 million agent–user interactions
a week powered by CopilotKit
1.200,000
1.000,000
800,000
400,000
200,000
0
1M+

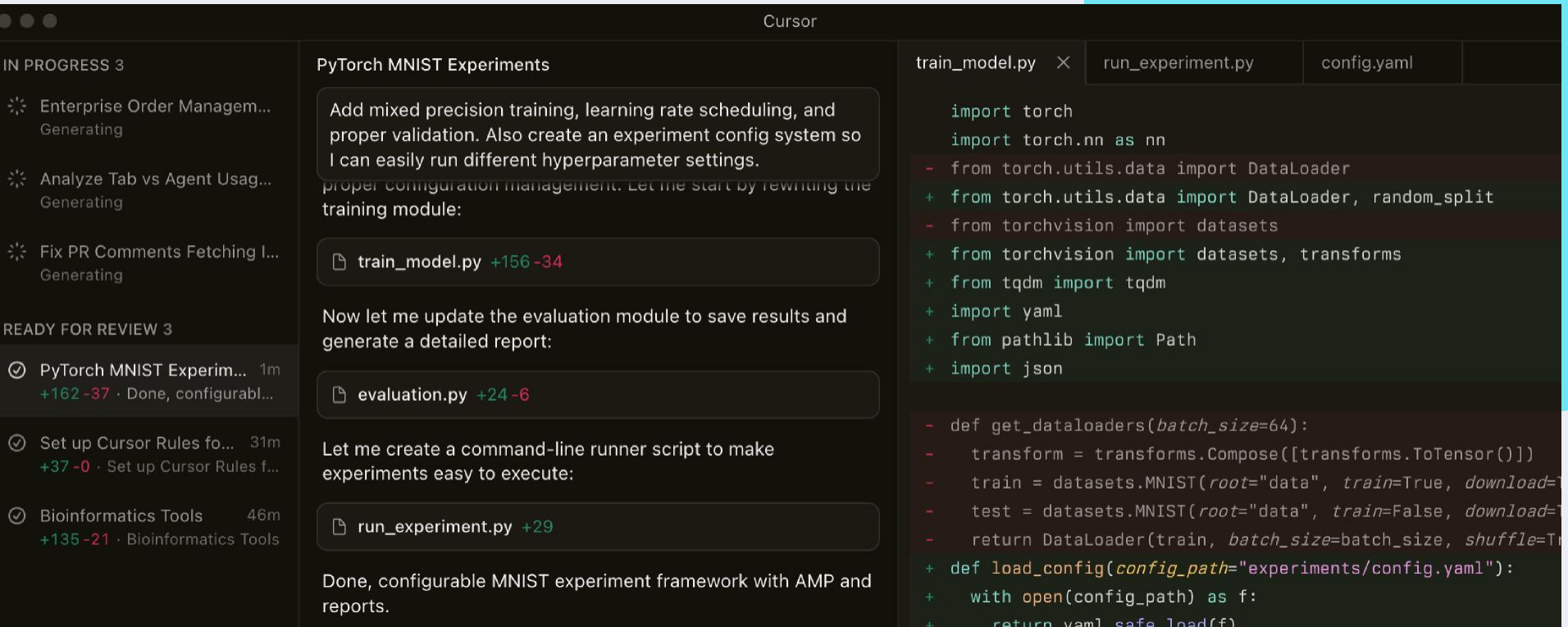
CopilotKit

CopilotKit


SaaS
Copilots
Productivity
Copilots

CopilotKit


Search HubSpot
Assistant
CopilotKit
Bookmarks
CRM
Marketing
Content
Sales
Commerce
Service
Data Management
Automation
Reporting
Add assistant
Breeze
Assistant
Ask me anything.
Add apps


Cursor
IN PROGRESS 3
Enterprise Order Managem...
Generating
Analyze Tab vs Agent Usag...
Generating
Fix PR Comments Fetching I...
Generating
READY FOR REVIEW 3
PyTorch MNIST Experim... 1m
+162 -37 · Done, configurabl...
Set up Cursor Rules fo... 31m
+37 -0 · Set up Cursor Rules f...
Bioinformatics Tools 46m
+135 -21 · Bioinformatics Tools
PyTorch MNIST Experiments
Add mixed precision training, learning rate scheduling, and proper validation. Also create an experiment config system so I can easily run different hyperparameter settings.
proper configuration management. Let me start by rewriting the training module:
train_model.py +156 -34
Now let me update the evaluation module to save results and generate a detailed report:
evaluation.py +24 -6
Let me create a command-line runner script to make experiments easy to execute:
run_experiment.py +29
Done, configurable MNIST experiment framework with AMP and reports.
train_model.py
run_experiment.py
config.yaml
import torch
import torch.nn as nn
- from torch.utils.data import DataLoader
+ from torch.utils.data import DataLoader, random_split
- from torchvision import datasets
+ from torchvision import datasets, transforms
+ from tqdm import tqdm
+ import yaml
+ from pathlib import Path
+ import json
- def get_dataloaders(batch_size=64):
- transform = transforms.Compose([transforms.ToTensor()])
- train = datasets.MNIST(root="data", train=True, download=
- test = datasets.MNIST(root="data", train=False, download=
- return DataLoader(train, batch_size=batch_size, shuffle=Tr
+ def load_config(config_path="experiments/config.yaml"):
+ with open(config_path) as f:

CopilotKit

The past 30 years of the internet

Request

↓

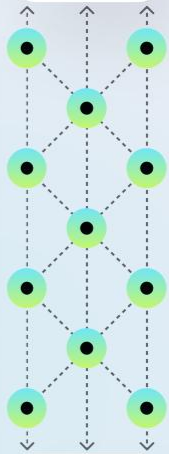
Response

Agentic Software

- Long running (require streaming)
- Structured + Unstructured I/O (text & voice + tool calls & state updates...)
- Require composition (hand-off control to subagents)
- Nondeterministic UI (inversion of control)

And more

AGENTS

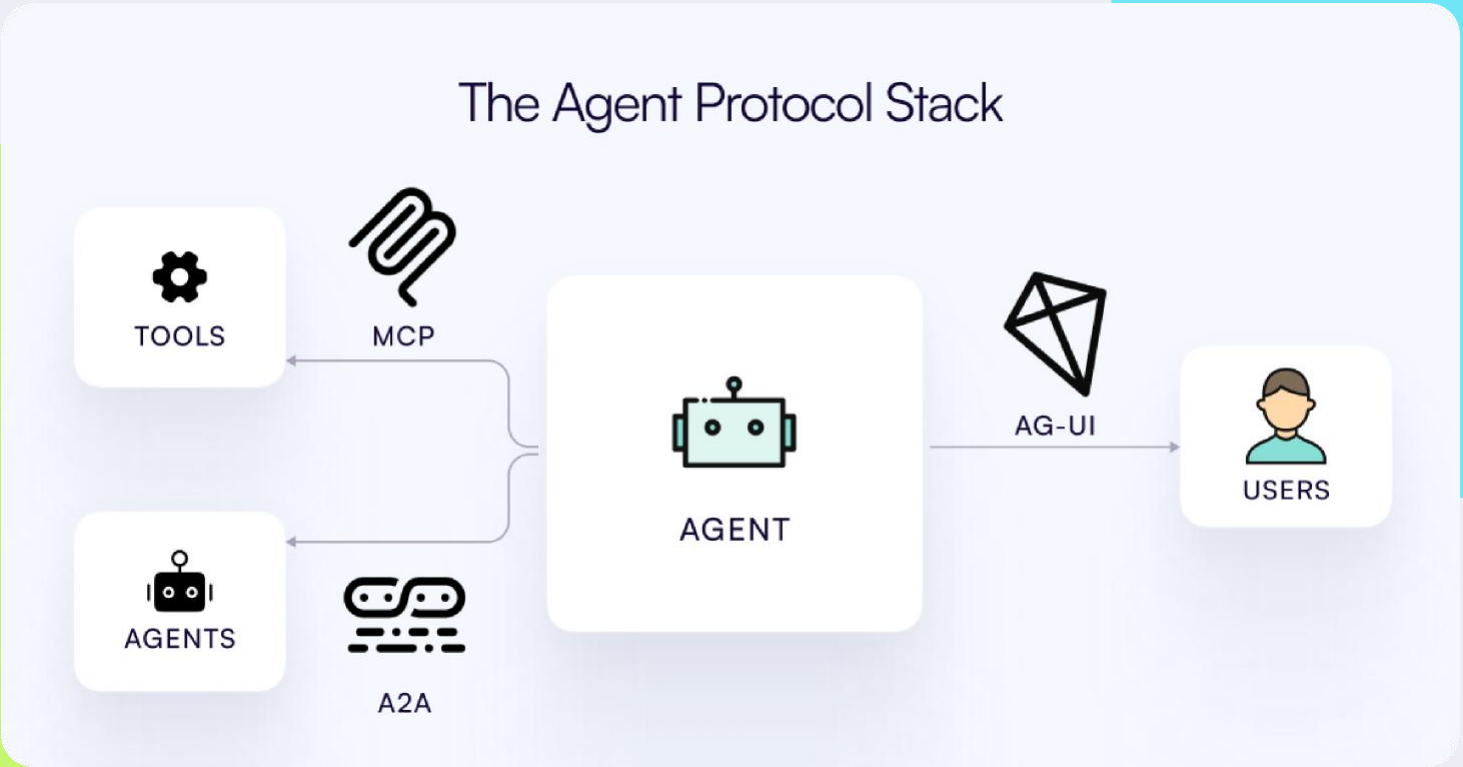


USER

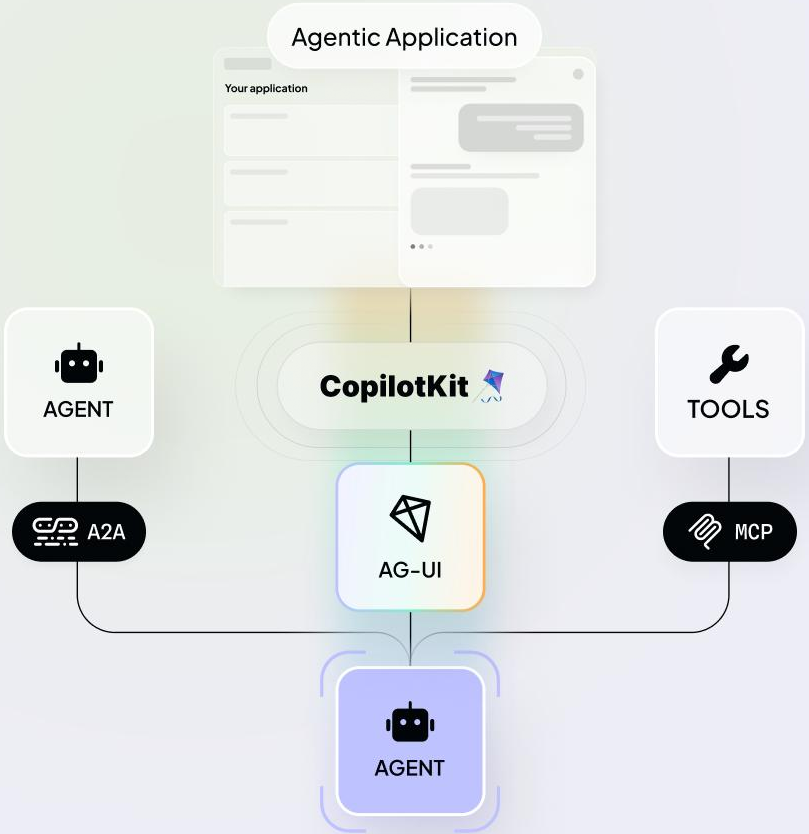
CopilotKit

CopilotKit

CopilotKit


The Agent Protocol Stack
TOOLS
MCP
AGENTS
A2A
AGENT
AG-UI
USERS

CopilotKit & AG-UI Connect Agents and Applications

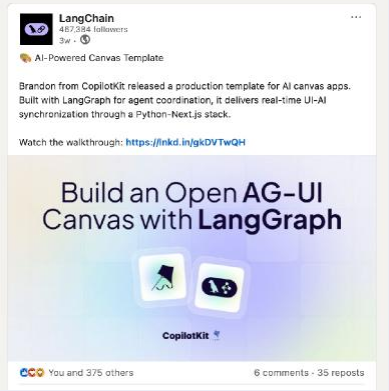


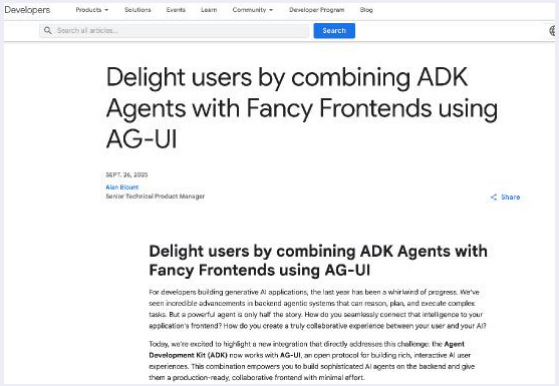
AG-UI

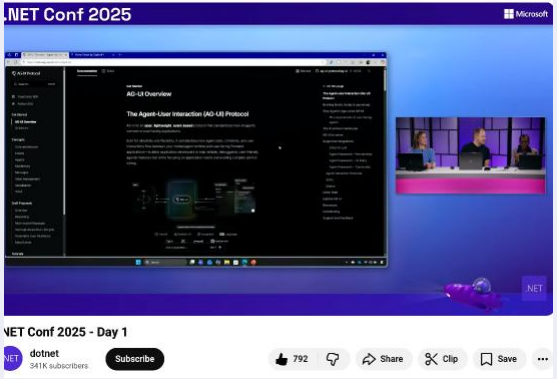
Event-based protocol connecting agentic backends & agentic frontends.

CopilotKit

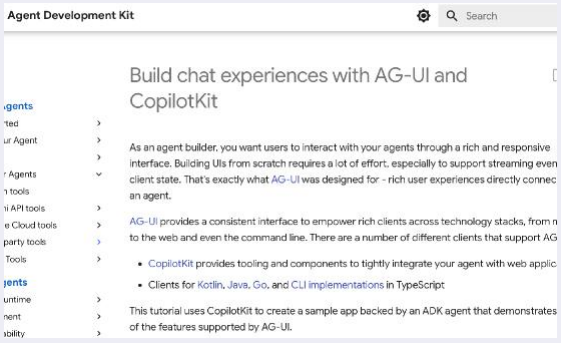
SDKs & Self-Hostable Platform for building agentic applications - purpose built for the agentic paradigm.

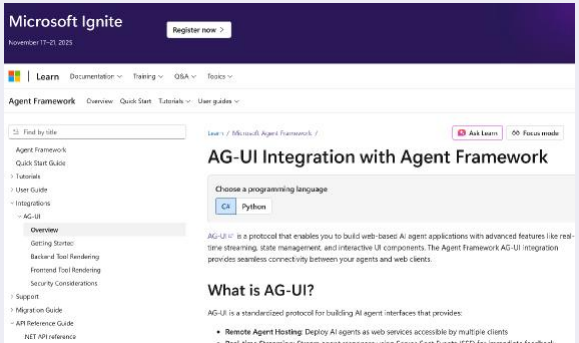

LangChain
487,384 followers
3w
AI-Powered Canvas Template
Brandon from CopilotKit released a production template for AI canvas apps. Built with LangGraph for agent coordination, it delivers real-time UI-AI synchronization through a Python-Next.js stack.
Watch the walkthrough: https://lnkd.in/gkDVTwQH
Build an Open AG-UI Canvas with LangGraph
CopilotKit
You and 375 others
6 comments · 35 reposts

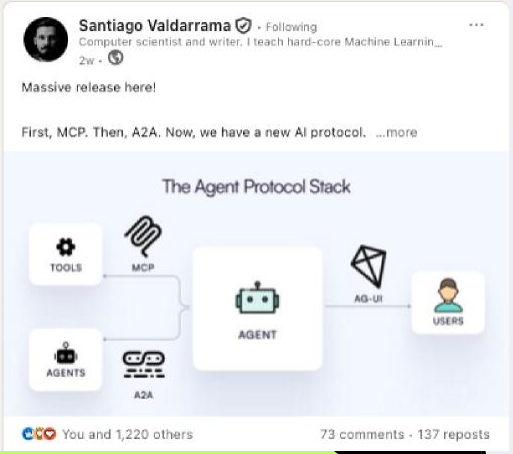

Developers
Products
Solutions
Events
Learn
Community
Developer Program
Blog
Search all articles...
Search
Delight users by combining ADK Agents with Fancy Frontends using AG-UI
SEPT. 24, 2025
Alan Blount
Senior Technical Product Manager
Share
Delight users by combining ADK Agents with Fancy Frontends using AG-UI
For developers building generative AI applications, the last year has been a whirlwind of progress. We've seen incredible advancements in backend agentic systems that can reason, plan, and execute complex tasks. But a powerful agent is only half the story. How do you seamlessly connect that intelligence to your application's frontend? How do you create a truly collaborative experience between your user and your AI?
Today, we're excited to highlight a new integration that directly addresses this challenge: the Agent Development Kit (ADK) now works with AG-UI, an open protocol for building rich, interactive AI user experiences. This combination empowers you to build sophisticated AI agents on the backend and give them a production-ready, collaborative frontend with minimal effort.

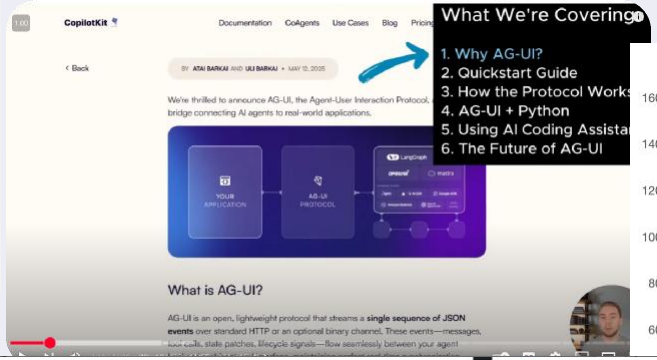

.NET Conf 2025
Microsoft
AG-UI Overview
The Agent-User Interaction (AG-UI) Protocol
.NET
.NET Conf 2025 - Day 1
dotnet
341K subscribers
Subscribe
792
Share
Clip
Save

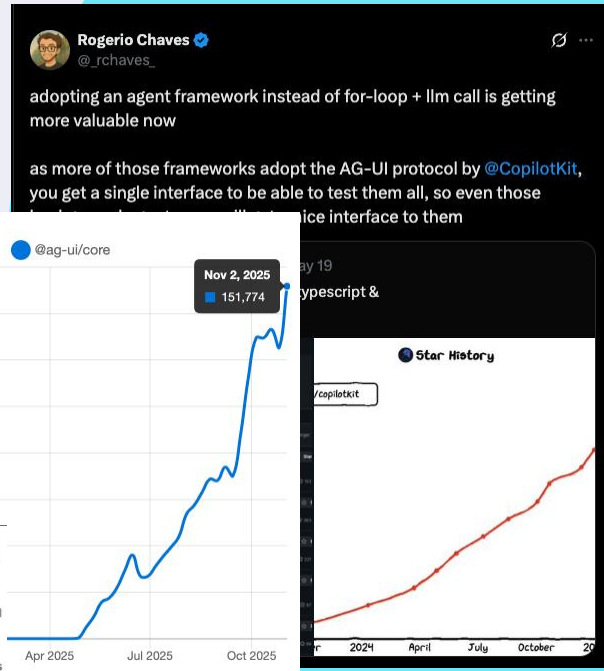

Building agentic copilots with CopilotKit and Mastra
Sam Bhagwat · Sep 18, 2025
Mastra is an agentic backend: it provides you the primitives like agents and workflows, to create specialized agents that can react.
But every backend needs a great agentic frontend.
One of the frontends we recommend is CopilotKit. CopilotKit provides interactive React components that hook into the Mastra streaming protocol and let you customize the user experience.
CopilotKit
mastra


Agent Development Kit
Search
Build chat experiences with AG-UI and CopilotKit
As an agent builder, you want users to interact with your agents through a rich and responsive interface. Building UIs from scratch requires a lot of effort, especially to support streaming even
client state. That's exactly what AG-UI was designed for - rich user experiences directly connec
an agent.
AG-UI provides a consistent interface to empower rich clients across technology stacks, from n
to the web and even the command line. There are a number of different clients that support AG
CopilotKit provides tooling and components to tightly integrate your agent with web applic
Clients for Kotlin, Java, Go, and CLI implementations in TypeScript
This tutorial uses CopilotKit to create a sample app backed by an ADK agent that demonstrates
of the features supported by AG-UI.

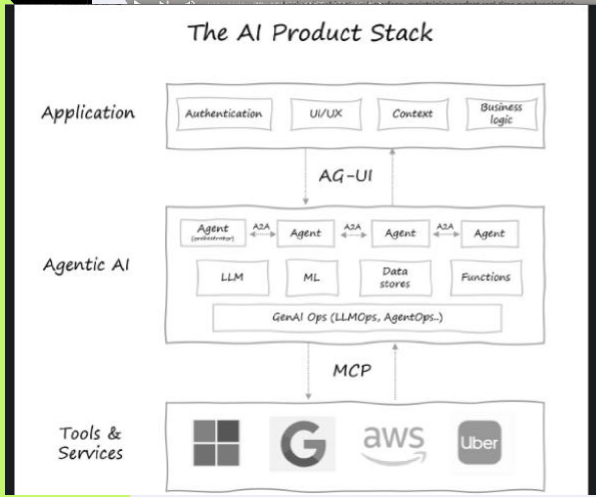

Microsoft Ignite
November 17–21, 2025
Register now
Learn
Documentation
Training
Q&A
Tools
Agent Framework
Overview
Quick Start
Tutorials
User guides
Ask Learn
Focus mode
AG-UI Integration with Agent Framework
Choose a programming language
C#
Python
What is AG-UI?
AG-UI is a standardized protocol for building AI agent interfaces that provides:
Remote Agent Hosting: Deploy AI agents as web services accessible by multiple clients


Santiago Valdarrama · Following
Computer scientist and writer. I teach hard-core Machine Learnin...
2w ·
Massive release here!
First, MCP. Then, A2A. Now, we have a new AI protocol. ...more
The Agent Protocol Stack
TOOLS
MCP
AGENTS
A2A
AGENT
AG-UI
USERS
You and 1,220 others
73 comments · 137 reposts


CopilotKit
Documentation
CoAgents
Use Cases
Blog
What We're Covering
1. Why AG-UI?
2. Quickstart Guide
3. How the Protocol Works
4. AG-UI + Python
5. Using AI Coding Assista
6. The Future of AG-UI
What is AG-UI?


Rogerio Chaves
@_rchaves_
adopting an agent framework instead of for-loop + llm call is getting more valuable now
as more of those frameworks adopt the AG-UI protocol by @CopilotKit, you get a single interface to be able to test them all, so even those
ice interface to them
ay 19
ypescript &
Star History
/copilotkit
2024
April
July
October

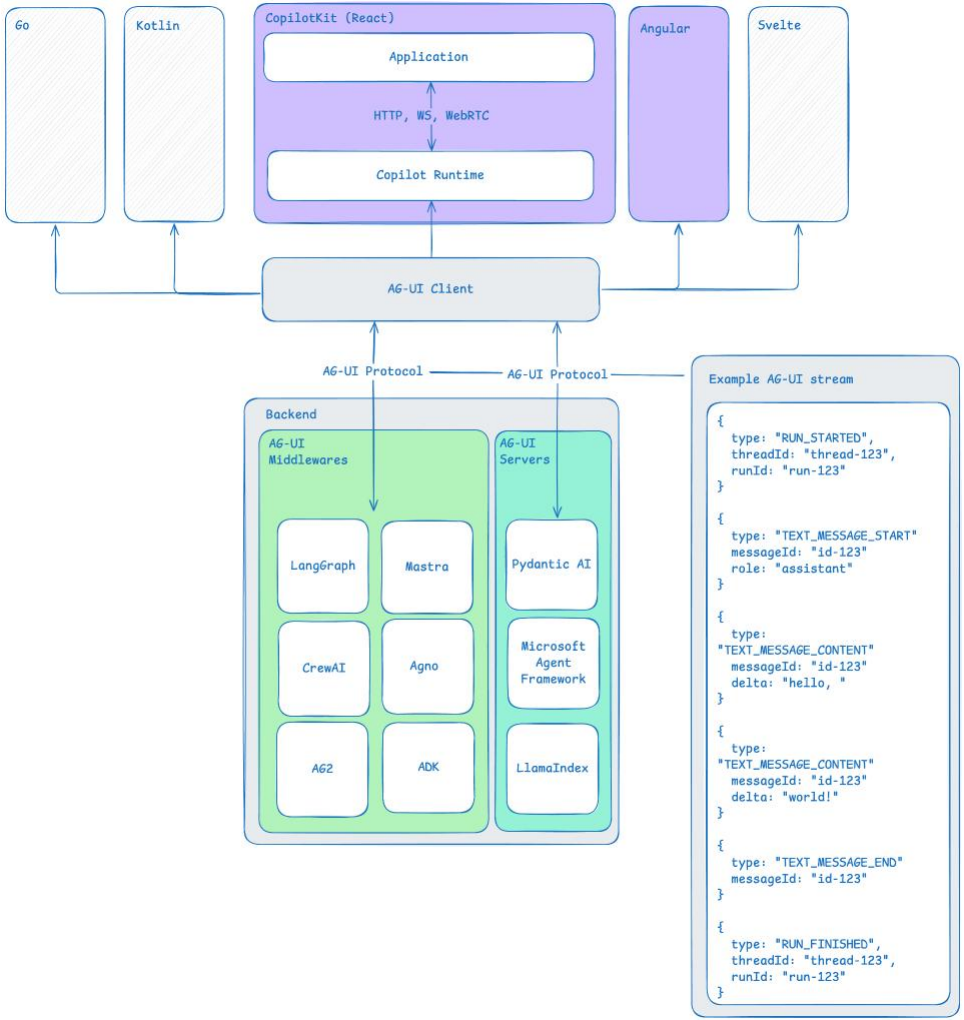
@ag-ui/core
Nov 2, 2025
151,774
160,000
140,000
120,000
100,000
80,000
60,000
Apr 2025
Jul 2025
Oct 2025


The AI Product Stack
Application
Authentication
UI/UX
Context
Business logic
AG-UI
Agentic AI
Agent
A2A
Agent
A2A
Agent
A2A
Agent
LLM
ML
Data stores
Functions
GenAI Ops (LLMOps, AgentOps..)
MCP
Tools & Services
aws
Uber


Amit Ghosh · 3rd+
AI Product Management @ M...
1w · Edited ·
+ Follow
Agentic AI will soon become the mainstream middleware and protocols such as MCP and AG-UI will unlock new paradigms of interoperability. The Tech product stack will start to look different with system boundaries emerging based on functionality and areas of specializations.
The application layer will be built by product managers and designers who'd be able to iterate and experiment quickly on the ideal user experience, using coding tools. Protocols such as AG-UI will establish the handshake with the Agentic AI workflows, which will form the core of the stack. Orchestration, multi-agent architecture and supporting data stores & functions will enable "thinking" and "reasoning", and GenAIOps with all its components (LLMOps, AgentOPs, FMOps) will hold it together at scale. Finally MCP will drive integration with enterprise / consumer tools and services for the end-to-end accomplishment of the jobs to be done.
728
37 comments · 81 reposts
Love
Comment
Repost
Send


Atai Barkai · 1st
CEO at CopilotKit.ai
2w ·
Introducing AG-UI;
The Agent-User Interaction protocol
...more
Agent User Interaction Protocol
Your application
Application User
Agent User Interaction Protocol
CopilotKit
The Agent Protocol Stack
AGENTS
TOOLS
A2A
AGENT
MCP

-
-
-


Go
Kotlin
CopilotKit (React)
Application
HTTP, WS, WebRTC
Copilot Runtime
Angular
Svelte
AG-UI Client
AG-UI Protocol
AG-UI Protocol
Backend
AG-UI Middlewares
LangGraph
Mastra
CrewAI
Agno
AG2
ADK
AG-UI Servers
Pydantic AI
Microsoft Agent Framework
LlamaIndex
Example AG-UI stream
{ type: "RUN_STARTED", threadId: "thread-123", runId: "run-123" }
{ type: "TEXT_MESSAGE_START" messageId: "id-123" role: "assistant" }
{ type: "TEXT_MESSAGE_CONTENT" messageId: "id-123" delta: "hello, " }
{ type: "TEXT_MESSAGE_CONTENT" messageId: "id-123" delta: "world!" }
{ type: "TEXT_MESSAGE_END" messageId: "id-123" }
{ type: "RUN_FINISHED", threadId: "thread-123", runId: "run-123" }

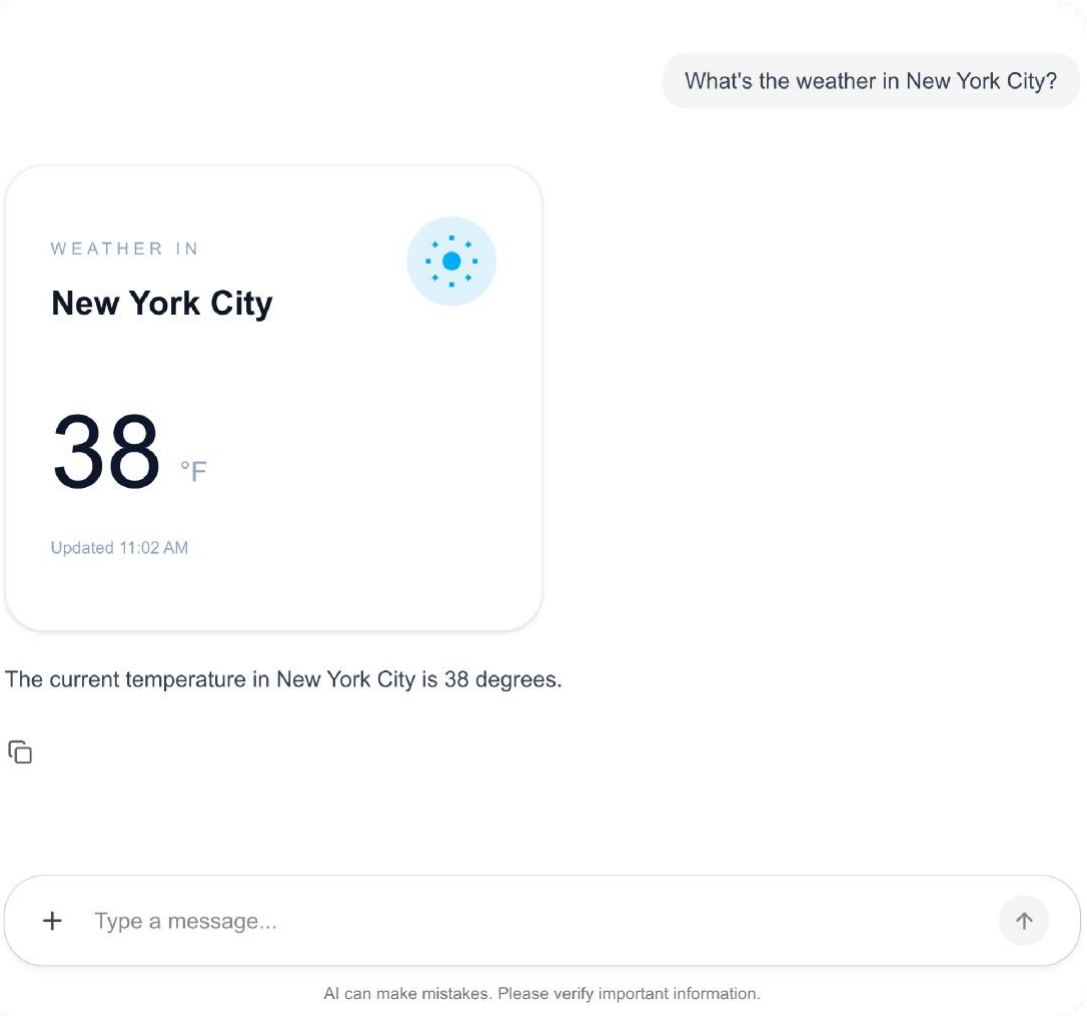
-
-
-

The current temperature in New York City is 38 degrees.

```python
# Agent

def get_weather(location: str):
  return {temp: 38, humidity: 50}

agent = agent(
    "gpt-5.1",
    tools=[get_weather]
)

agent.ag_ui.serve()
```

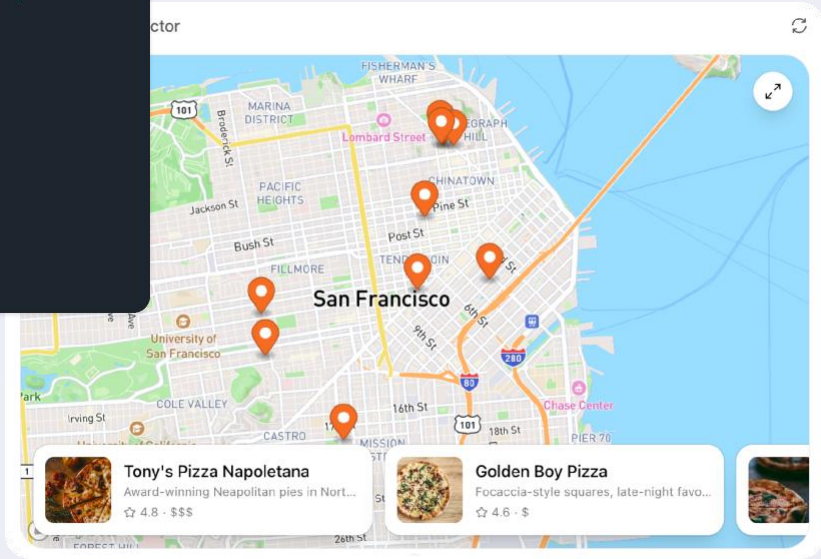
```tsx
// Frontend

import {
  CopilotChat
  useRenderTool
} from "@copilotkit/react-ui/v2"

useRenderTool({
  name: "get_weather",
  render: ({ args, result }) => {
    return <WeatherCard
      location={args.location}
      temp={result.temp}
    />;
  },
});

return <CopilotChat/>
```

```tsx
// Frontend

import {
  CopilotChat
  useRenderTool
} from "@copilotkit/react-ui/v2"

useRenderTool({
  name: "get_weather",
  render: ({ args, result }) => {
    return <WeatherCard
      location={args.location}
      temp={result.temp}
    />;
  },
});

return <CopilotChat/>
```

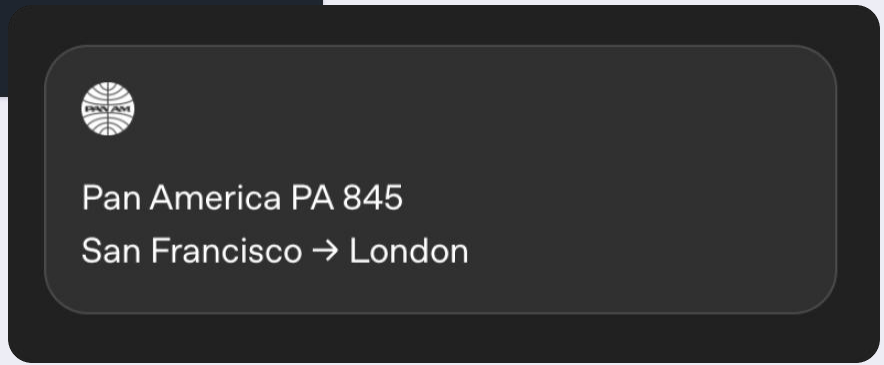


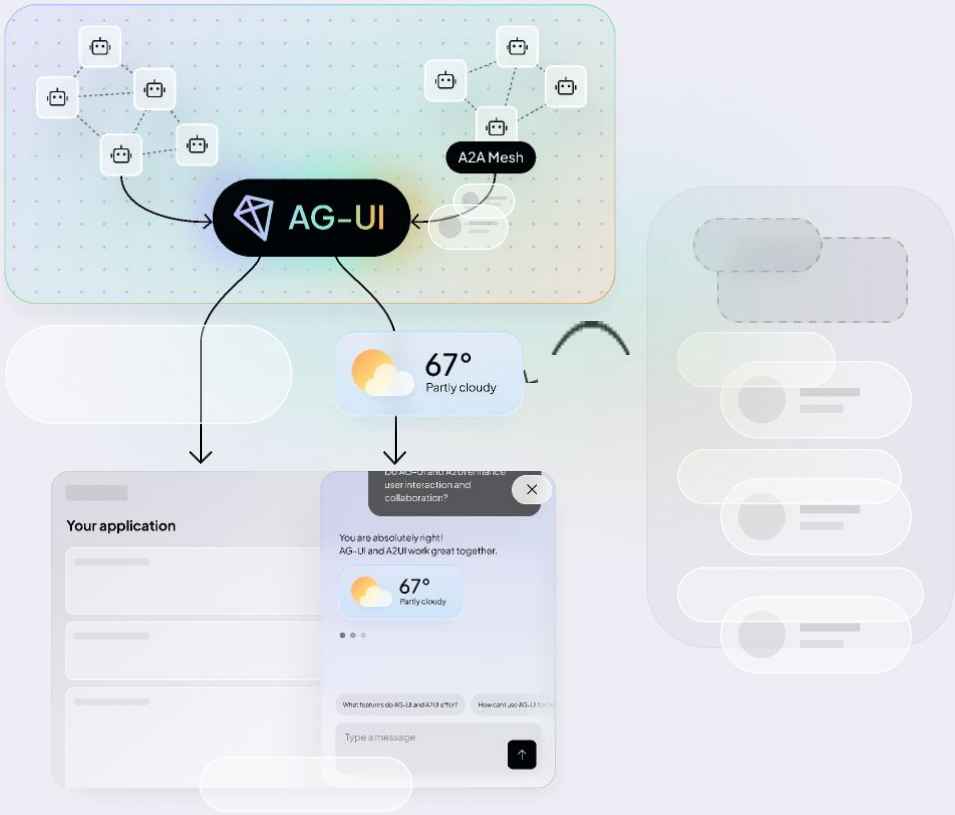
```tsx
// Frontend
const { agent } = useAgent()

return (
  <div>
    {agent.messages.map(
      (message) => <div>
        { message.content }
        { message.ui() }
      </div>
    }
  </div>
)
```

```html
<!-- ChatGPT UI -->
<html>
  <body>
    <!-- ChatGPT wraps your app in sandboxed iframes -->
    <iframe src="https://web-sandbox.oaiusercontent.com/...">
      <iframe src="https://your-app.example.com/">
        <!-- Your ChatGPT App SDK UI runs here -->
      </iframe>
    </iframe>
  </body>
</html>

```



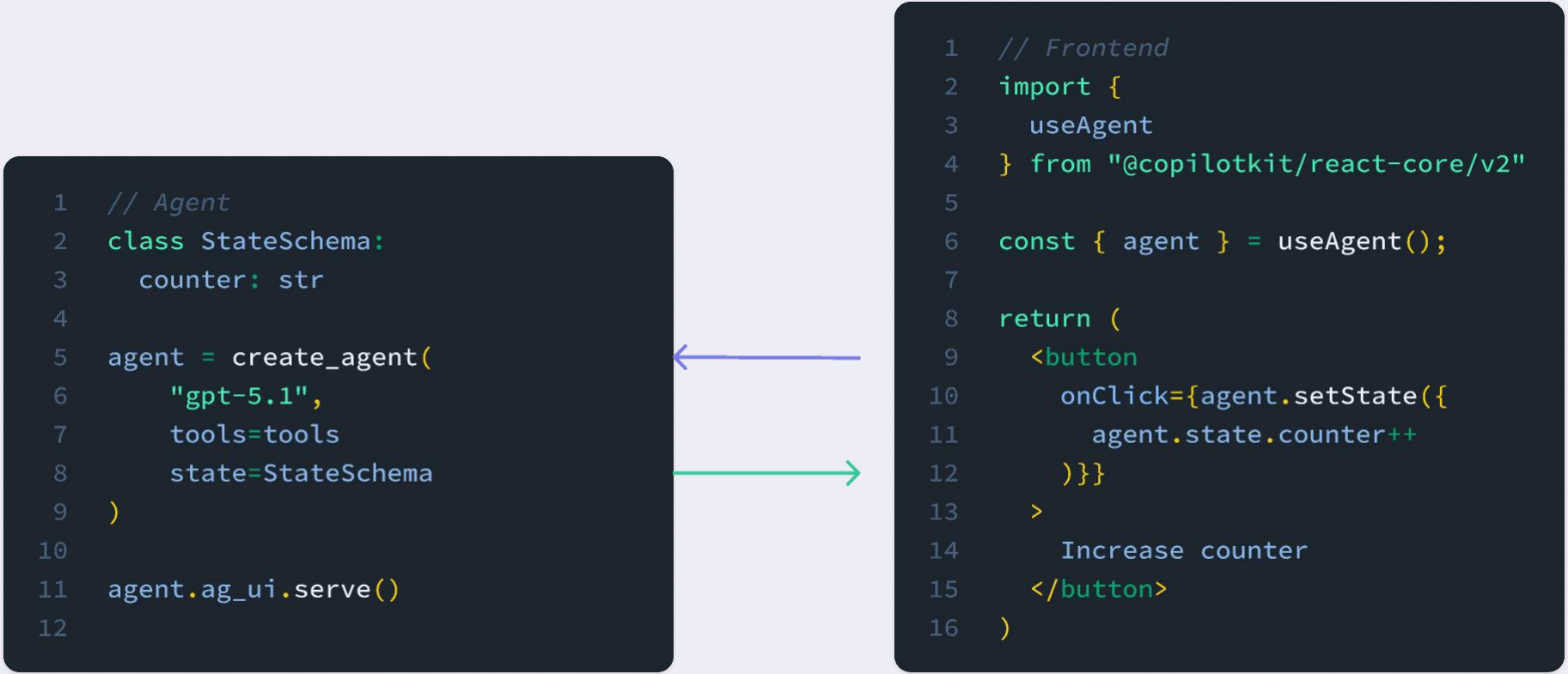
```
<Card>
  <Image src={airline.logo} size={24} />

  <Text value={`${airline.name} ${number}`} />

  <Text value={`${departure.city} → ${arrival.city}`} />
</Card>

```



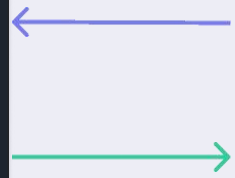

A2A Mesh
AG-UI
67°
Partly cloudy
Your application
user interaction and
collaboration?
You are absolutely right!
AG-UI and A2UI work great together.
67°
Partly cloudy
Type a message

```python
// Agent
class StateSchema:
  name: str

agent = create_agent(
    "gpt-5.1",
    tools=tools
    state=StateSchema
)

agent.ag_ui.serve()

```

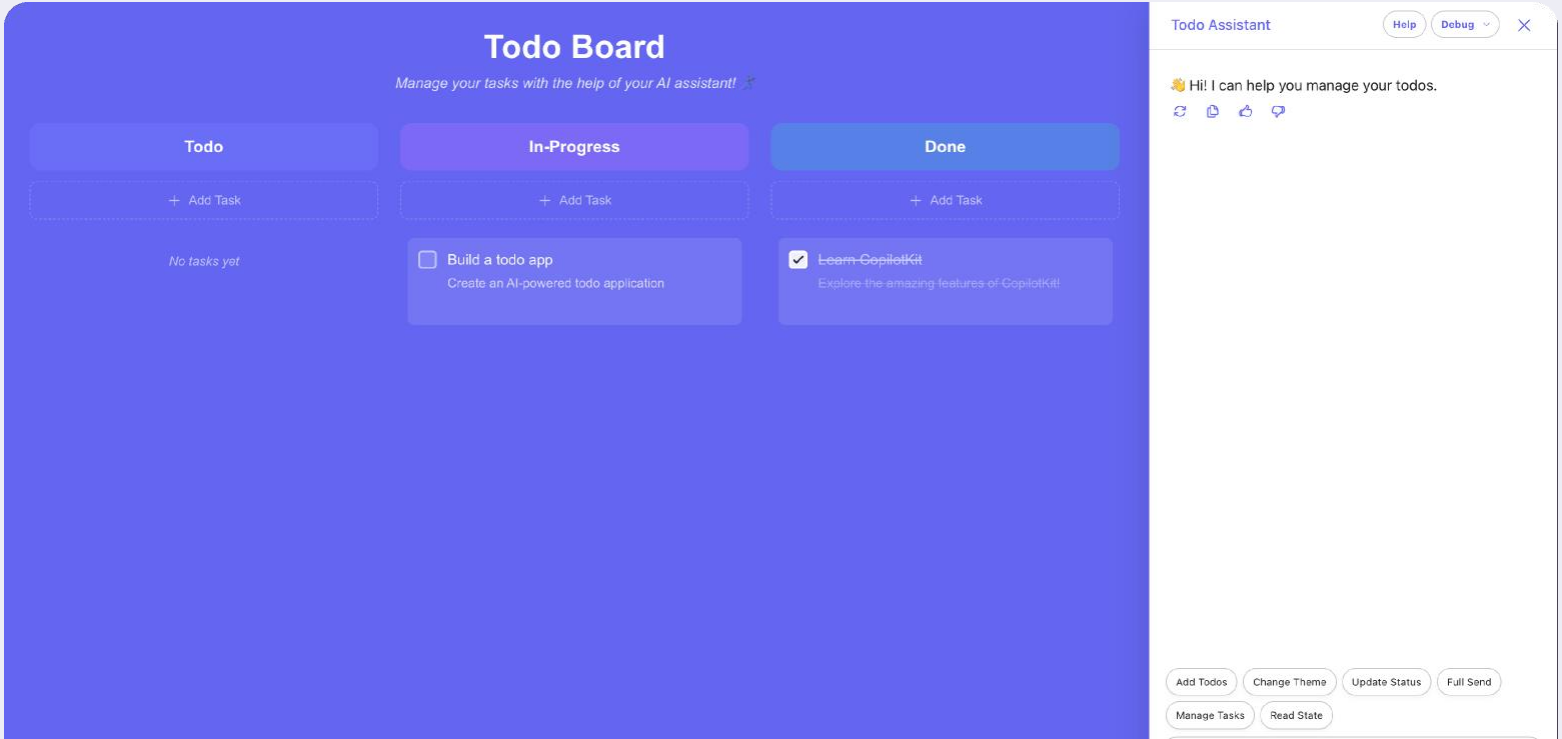
```tsx
// Frontend
import {
  useAgent
} from "@copilotkit/react-core/v2"

const { agent } = useAgent();

return (
  <div>
    { agent.state.name }
  </div>
)
```


// Agent
class StateSchema:
counter: str
agent = create_agent(
"gpt-5.1",
tools=tools
state=StateSchema
)
agent.ag_ui.serve()
// Frontend
import {
useAgent
} from "@copilotkit/react-core/v2"
const { agent } = useAgent();
return (
<button
onClick={agent.setState({
agent.state.counter++
)}}
>
Increase counter
</button>
)

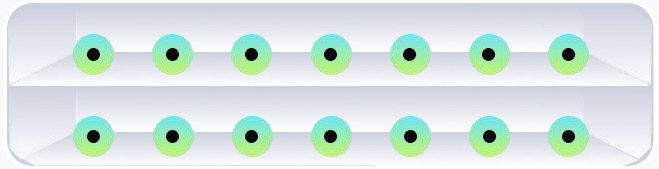
```python
# Agent
class StateSchema:
  document: str

agent = create_agent(
    "gpt-5.1",
    tools=[generate_document]
    state=StateSchema
)

agent.ag_ui.serve()

```

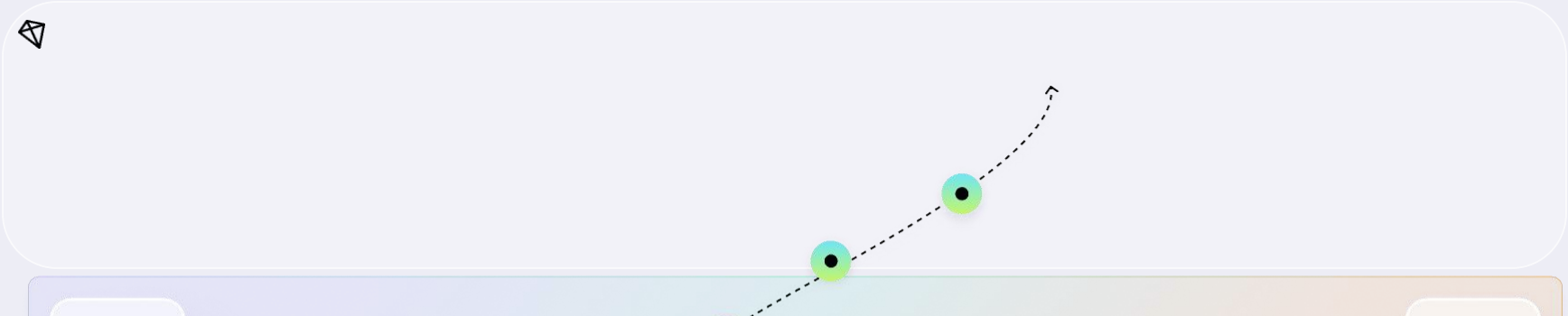


```tsx
// Frontend
import {
  useAgent
} from "@copilotkit/react-core/v2"

const { agent } = useAgent();
const [changes, setChanges] = ...

return (
  <main>
    <p>{agent.document}</p>
    <button
      onClick={agent.setState({
        document: changes
      )}}
    >
      Save changes
    </button>
  </main>
)

```


Todo Board
Manage your tasks with the help of your AI assistant!
Todo
In-Progress
Done
Add Task
Add Task
Add Task
No tasks yet
Build a todo app
Create an AI-powered todo application
Learn CopilotKit
Explore the amazing features of CopilotKit!
Todo Assistant
Help
Debug
Hi! I can help you manage your todos.
Add Todos
Change Theme
Update Status
Full Send
Manage Tasks
Read State

-
-
-





CopilotKit

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AG-UI: The Agent-User Interaction Protocol

AG-UI is an open, lightweight, event-based protocol that standardizes how AI agents connect to user-facing applications. Built for simplicity and flexibility, it enables seamless integration between AI agents, real time user context, and user interfaces.

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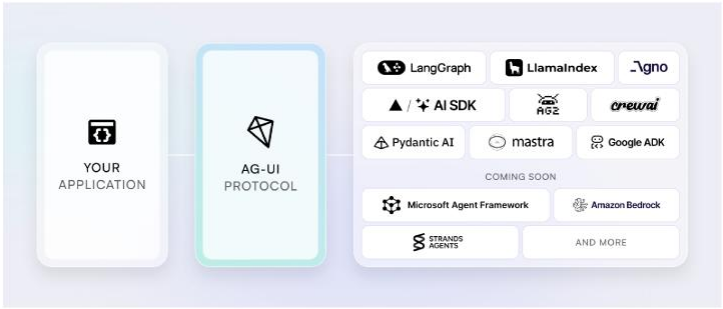

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